

The complementary-distribution challenge for phonological learning: The case of [h] and [ɱ] in English

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Abstract

The distribution of [h] and [ɱ] in English poses a well-known challenge for the structuralist assumption that segments in complementary distribution are allophones of the same phoneme. In this paper, we show that the distribution of [h] and [ɱ] also poses a challenge for simplicity-based theories of phonological learning. Under naive representational assumptions, simplicity-based learners prefer to derive [ɱ] from an underlying /h/, rather than a longer sequence of /ng/, and therefore make wrong predictions about English speakers' knowledge. However, when combined with a representational framework assuming underspecification and lexical redundancy rules, the same learners make the correct predictions. We therefore propose that the complementary distribution challenge does not constitute an argument against simplicity-based learning, but an argument in favor of underspecification and lexical redundancy rules.

1 Introduction

The consonants [h] and [ɱ] are in complementary distribution in English, a fact that has puzzled phonologists for a century. Twaddell (1935, p. 29) writes:

A perfectly consistent and unambiguous transcription of English is possible with the use of a single symbol in the place of [h] and [ɱ]; but no phonetician or phonologist could consent to regarding these two ‘sounds’ as variants of the same phoneme.

*Donca Steriade has pointed out to us the challenge posed by the complementary distribution of [h] and [ɱ] at the MISTI workshops on abstractness in phonology that took place at Tel-Aviv University in January 2023 and MIT in February 2025. Steriade has also provided us with the grammars in section 2. We thank her, Adam Albright, and all of the participants in the workshops, especially to Maya Nitzan who worked with us on the project in its early stages. We also thank Snezana Iovtcheva for her help with Bulgarian.

This observation poses a well-known challenge for the structuralist phonemic method, according to which segments in complementary distribution belong to the same phoneme.

As we discuss in this paper, however, the distribution of [h] and [ŋ] also poses an unfamiliar challenge for generative phonology. Specifically, a challenge from the perspective of simplicity-based phonological learning. We will spell out this challenge in sections 2 and 3 but for now, simplifying, there are at least two observationally adequate grammars of English that are consistent with the distribution of [h] and [ŋ]: the descriptively correct grammar of English, which derives [ŋ] from underlying /ng/, and a descriptively incorrect grammar that derives [ŋ] from /h/ in coda position. The challenge is to achieve explanatory adequacy with respect to the descriptively correct grammar.¹

The challenge is raised by the simpler lexical representations of the incorrect /h/ → [ŋ] grammar. In this grammar, words like ‘song’ can be represented with one fewer segment than equivalent words in the correct grammar (e.g., /sɒh/ instead of /sɒng/). As will be shown in section 3, under naive representational assumptions, a family of simplicity-based learners favor the descriptively incorrect option. These are learners that prefer the acquisition of the simplest observationally adequate grammar, where grammar simplicity takes into consideration the information in the lexicon and phonological rules/constraints, as in the Sound Pattern of English (SPE; Chomsky and Halle 1968). We will use the term SIMPLICITY, as defined in (1), to refer to all relevant simplicity-based learning approaches.

- (1) SIMPLICITY: A learning theory according to which the learner chooses the simplest observationally adequate grammar, all other things being equal.

The /h/ → [ŋ] challenge for SIMPLICITY dates back to Chomsky and Halle’s (1968) original SPE metric. This challenge has never been discussed in the literature and could have been set aside for decades because SIMPLICITY was largely abandoned within phonology. However, SIMPLICITY has recently returned in the form of Minimum Description Length (MDL; Rissanen 1978). MDL is a simplicity-based learner that has been implemented with demonstrated success and, as we will discuss in section 3, can support the acquisition of phonological grammars and morphological segmentation from distributional evidence. The /h/ → [ŋ] challenge is, to our knowledge, the first known empirical challenge for the MDL learner. We therefore believe that it cannot be ignored.

As we will show in section 5, however, SIMPLICITY’s choice of the /h/ → [ŋ] grammar only holds assuming full specification and Richness of the Base (Prince and Smolensky, 1993/2004; Smolensky, 1996). The simplicity-based learning criterion is not tied to a specific representational theory, and different choices about the architecture of grammar can affect its predictions regarding learning. As we will see, SIMPLICITY can make the correct predictions when combined with Chomsky & Halle’s (1968) architecture, which assumes underspecification and lexical redundancy rules. A summary of our observation is shown in (2).

¹We use Chomsky’s (1965) levels of adequacy: ‘*Observationally adequate*’ as a term describing consistency with the data; ‘*Descriptively adequate*’ means that the grammar reflects speakers’ knowledge; ‘*Explanatory adequate*’ means that the theory can correctly predict the acquisition of the adult grammar.

- (2) A summary of our observation from section 5

Representational assumptions	SIMPLICITY's prediction regarding learning
Full specification	English speakers will learn the /h/ → [ɰ] grammar
Underspecification + Lexical redundancy rules	English speakers will learn the /ng/ → [ɰ] grammar

We will discuss this table in detail in section 5, but here is the logic behind it in a nutshell. The incorrect /h/ → [ɰ] grammar includes underlying /h/ in coda position (e.g, /soh/ → [soɰ] ‘song’). Consequently, this grammar misses a generalization that otherwise holds for the English lexicon, according to which there are no /h/’s in coda positions whatsoever. The /h/ → [ɰ] grammar therefore needs to store in the lexicon enough features to distinguish between coda /h/ and coda /k/ (to distinguish between words like ‘song’ and ‘sock’ for example). This is illustrated in (3), where the feature [±continuant] is required. The correct /ng/ → [ɰ] grammar, however, does not miss the generalization regarding the exclusion of /h/ from coda position in the lexicon. Therefore, in this grammar it is redundant to specify the feature [-continuant] for coda /k/, as illustrated in (4). Under a theory with underspecification and lexical redundancy rules, this feature can be removed from all /k/’s in coda position across the lexicon.

- (3) Underspecified segments in coda position, **incorrect grammar**

	p	b	...	k	h
±coronal	-	-	...	-	-
±anterior	+	+	...	-	-
...			...		
±voice	-	+	...	-	-
±continuant	-	-	...	-	+

- (4) Underspecified segments in coda position, **correct grammar**

	p	b	...	k
±coronal	-	-	...	-
±anterior	+	+	...	-
...			...	
±voice	-	+	...	-
±continuant	-	-	...	

As we will see, the elimination of this feature from the lexicon in the correct /ng/ → [ɰ] grammar is more cost-effective than the simplification of lexical representations in the incorrect /h/ → [ɰ] grammar. The reasoning, presented more elaborately in section 5, is the following. As long as segments can be underspecified, [k] in coda position is cheaper in the correct grammar, but [ɰ] is cheaper in the incorrect one. Because there are sufficiently more coda [k]’s than [ɰ]’s, SIMPLICITY favors the correct grammar. If this solution is on the right track, this would mean that the complementary-distribution challenge is not an argument against SIMPLICITY, but an argument in favor of the representational assumptions in (5). According to these assumptions, segments can be minimally-specified in any context across the lexicon, as long as each segment is still uniquely specified. The remaining features, which are redundant for the purpose of distinctness, are filled in by phonological rules.

- (5) Representational assumptions that can solve the learning challenge

1. Underlying segments can be underspecified for phonological features as long as they are pairwise distinct in every environment.
2. Unspecified features are filled in by lexical redundancy rules.

This paper is structured as follows. In section 2, we outline two observationally adequate grammars of English: one grammar in which [ŋ] is derived from /ng/ and a second grammar in which [ŋ] is derived from /h/. In section 3, we show by calculation that under naive representational assumptions, SIMPLICITY favors the descriptively incorrect grammar. In section 4, we discuss some properties of this case study that make the /h/ → [ŋ] challenge an instance of a general problem for phonological learning. The generality of the problem is significant for our choice of solution to follow. In section 5, we will present the solution just outlined, which is based on underspecification and lexical redundancy rules. Section 6 concludes.

2 Two observationally adequate grammars

2.1 /ng/ → [ŋ] grammar (correct)

In English, both [h] & [ŋ] and [n] & [ŋ] are in complementary distribution. The environments in which all three segments surface are shown in (6).

- (6) The complementary distributions of [h] & [ŋ] and [n] & [ŋ] in English (cf. Smith 1982; Steriade 2023; C represents any consonant other than [g] and [k])

	[h]	[ŋ]	[n]
#__	✓	✗	✓
— $\begin{Bmatrix} g \\ k \end{Bmatrix}$	✗	✓	only as a part of the prefix <i>un-</i>
__C	✗	only preceding a deleted underlying /g/	✓
__#	✗	only preceding a deleted underlying /g/	✓
V__V	✓	only preceding a deleted underlying /g/	✓

The distribution of the three segments in (6) licenses the construction of two observationally adequate grammars, which we will describe in this section. First, a grammar in which [ŋ] is only derived from /n/ before /g/ and /k/, and second, a grammar in which [ŋ] is also derived from /h/ in coda position. We will see that both grammars can generate the correct surface forms and are therefore observationally adequate. For clarity, grammars are written within the framework of rule-based phonology but the implications should hold for Optimality Theory (OT; Prince and Smolensky 1993/2004) as well.²

²As we will see in section 3, the shorter underlying representations of the /h/ → [ŋ] grammar are beneficial in terms of SIMPLICITY because they extensively outweigh the additional complexity introduced by an /h/

There is at least one further observationally adequate grammar to consider: an /ŋ/ → [ŋ] grammar. We will not outline this grammar because SIMPLICITY will favor the other two over it. This is due to a massive SIMPLICITY pressure to minimize the underlying inventory of a language. As will be explained in more detail later, the number of underlying segments determines the encoding cost of each segment in the lexicon. Therefore, adding a segment to the underlying inventory inflates the cost of *every* segment in *every* lexical item. The predictability of [ŋ], which can be derived from other underlying segments (/ng/ or /h/), rules out the possibility that SIMPLICITY will choose the /ŋ/ → [ŋ] grammar.

Let us start with the /ng/ → [ŋ] grammar. In this grammar, every [ŋ] is derived from an underlying /n/ that precedes an underlying /g/ or /k/, as can be seen in the mappings from underlying representations (URs) to surface representations (SRs) in (7). This assimilation pattern is seen transparently in (7a)-(7b) and opaquely in (7c)-(7d), where the trigger for assimilation (/g/) is dropped by a post-nasal consonant deletion process that targets voiced stops.

(7) /ng/ → [ŋ] lexicon and mappings

- | | |
|--|--|
| a. /in-kwést/ → [iŋkwést] ‘ <i>inquest</i> ’ | g. /omaha/ → [ómahà] ‘ <i>Omaha</i> ’ |
| b. /fingər/ → [fíŋgər] ‘ <i>finger</i> ’ | |
| c. /sing/ → [síŋ] ‘ <i>sing</i> ’ | h. /ɛks-heɪl/ → [ɛgzéɪl] ~ [ɛgzhéɪl] ‘ <i>exhale</i> ’ |
| d. /inglænd/ → [íŋlænd] ‘ <i>England</i> ’ | |
| e. /bɒmb/ → [bóm] ‘ <i>bomb</i> ’ | i. /vɪhɪkl/ → [víɪkl] ~ [víhɪkl] ‘ <i>vehicle</i> ’ |
| f. /hat/ → [hát] ‘ <i>hat</i> ’ | |

The example in (7e) shows that post-nasal stop deletion also applies to voiced labials, as in the mapping of /bɒmb/ → [bóm] (such a process does not apply to voiced coronals, e.g., [énd] ‘*end*’). All other examples show the mapping of an underlying /h/, which can undergo an optional h-dropping process in some environments, as shown in (7h)-(7i).

The phonological rules that account for the alternations in (7) are formalized in (8). The nasal-place assimilation rule is presented in (8a). The rule that deletes voiced dorsals and labials after nasals is presented in (8b). The optional rules in (8c)-(8d) account for the h-dropping in (7h) and (7i) respectively.

(8) /ng/ → [ŋ] phonological rules

- a. n → [α place] / __[α place]
- b. [-coronal, +voice] → ∅ / [+nasal]__σ
- c. h → ∅ / C__ (optional)
- d. h → ∅ / __V[-stress] (optional)

→ [ŋ] phonological rule. In OT as well, a cheaper lexicon will push the learner to make the same choice. If constraints are learned, the small counterweight of the rule will be translated into a similarly negligible counterweight of a constraint. If all constraints are innate, there is no counterweight against the lexicon to begin with, since all constraint rankings are equally simple.

The derivations in (9) show that the /ng/ → [ŋ] grammar is observationally adequate in the sense that the lexical representations in (7), when combined with the rules in (8), can lead to the correct surface forms.³

(9) /ng/ → [ŋ] derivations

UR	/in-kwɛst/	/fɪŋgər/	/sɪŋg/	/bɒmb/	/ɛks-heɪl/	/vɪhɪkl/
n → [α place] / __[α place]	ɪŋkwɛst	fɪŋgər	sɪŋg	-	-	-
[-coronal, +voice] → ∅ / [+nasal]__ _σ	-	-	sɪŋ	bɒm	-	-
h → ∅ / C__ (opt.)	-	-	-	-	ɛksɛɪl	-
h → ∅ / __V[-stress] (opt.)	-	-	-	-	-	vɪɪkl
SR	[ɪŋkwɛst]	[fɪŋgər]	[sɪŋ]	[bɒm]	[ɛgzɛɪl]	[vɪɪkl]

2.2 /h/ → [ŋ] grammar (incorrect)

We now turn to the /h/ → [ŋ] grammar. As can be seen in (10a), in this grammar as well, some [ŋ]'s that alternate with [n] must be derived from an underlying /n/. Since nasal place assimilation is necessary in this grammar as well, the [ŋ] in (10b) can be derived from either an /n/ or an /h/. This decision will have no effect on the evaluation by SIMPLICITY. The crucial difference between this grammar and the grammar in 2.1 is shown in (10c)-(10d) (in boldface). In those examples, the derivation of [ŋ] from an /h/ instead of a sequence of two consonants /ng/ serves an economical purpose.

(10) /h/ → [ŋ] lexicon and mappings

- | | |
|--|---|
| a. /in-kwɛst/ → [ɪŋkwɛst] ‘ <i>inquest</i> ’ | g. /omaha/ → [ómahà] ‘ <i>Omaha</i> ’ |
| b. /fɪŋgər/ → [fɪŋgər] ‘ <i>finger</i> ’ | |
| c. /sɪh/ → [sɪŋ] ‘ <i>sing</i> ’ | h. /ɛks-heɪl/ → [ɛgzɛɪl] ~ [ɛgzɛɪl] ‘ <i>exhale</i> ’ |
| d. /ɪhlænd/ → [ɪŋlænd] ‘ <i>England</i> ’ | |
| e. /bɒmb/ → [bɒm] ‘ <i>bomb</i> ’ | i. /vɪhɪkl/ → [vɪɪkl] ~ [vɪhɪkl] ‘ <i>vehicle</i> ’ |
| f. /hat/ → [hát] ‘ <i>hat</i> ’ | |

All other examples show no difference from the grammar in 2.1. As can be seen in (10e), post-nasal deletion in coda position must play a role in this grammar as well, even if [ŋ]'s in coda position are derived from /h/.

The phonological rules of this grammar are shown in (11). All four rules from (8) are required here as well. The only difference between the grammars in terms of rules is the additional rule (11e) (in boldface), which is responsible for the derivation of [ŋ] from /h/ in coda position.⁴

³The example in (7d) – [ɪŋlænd] – is derived in the same way as (7c) – [sɪŋ]. None of the rules in (8) can apply in the derivations of the examples in (7f)-(7g). They are therefore absent from (9) as well. For simplicity, we do not show additional rules that are not relevant to this paper, such as stress assignment and the voicing of the /ks/ sequence in /ɛks-heɪl/ → [ɛgzɛɪl].

⁴The rule in (11b) could be alternatively defined using [+labial, +voice], but this will not affect the discussion later on, because this change does not affect the complexity of the rule.

(11) /h/ → [ɥ] phonological rules

- a. $n \rightarrow [\alpha \text{ place}] / _ [\alpha \text{ place}]$
- b. $[-\text{coronal}, +\text{voice}] \rightarrow \emptyset / [+nasal] _]_{\sigma}$
- c. $h \rightarrow \emptyset / C _ (\text{optional})$
- d. $h \rightarrow \emptyset / _ V[-\text{stress}] (\text{optional})$
- e. **$h \rightarrow \eta / _]_{\sigma}$**

To conclude this section, the derivations in (12) show that the /h/ → [ɥ] grammar is observationally adequate as well. Therefore, this grammar competes with the /ng/ → [ɥ] grammar from the perspective of the learner, to which we turn next.

(12) /h/ → [ɥ] derivations

UR	/in-kwɛst/	/fɪŋgər/	/sih/	/bɒmb/	/ɛks-heɪl/	/viɦɪkl/
$n \rightarrow [\alpha \text{ place}] / _ [\alpha \text{ place}]$	ɪŋkwɛst	fɪŋgər	-	-	-	-
$[-\text{coronal}, +\text{voice}] \rightarrow \emptyset / [+nasal] _]_{\sigma}$	-	-	-	bɒm	-	-
$h \rightarrow \emptyset / C _ (\text{opt.})$	-	-	-	-	ɛkseɪl	-
$h \rightarrow \emptyset / _ V[-\text{stress}] (\text{opt.})$	-	-	-	-	-	viɦkl
$h \rightarrow \eta / _]_{\sigma}$	-	-	siŋ	-	-	-
SR	[ɪŋkwɛst]	[fɪŋgər]	[sɪŋ]	[bɒm]	[ɛgzɛɪl]	[viɦkl]

3 Descriptive adequacy and the challenge of explanatory adequacy

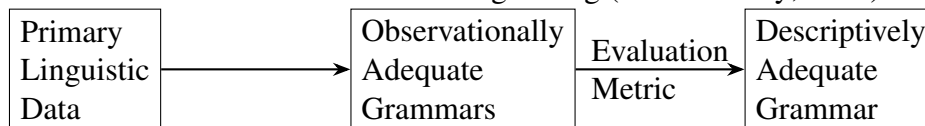
As shown in section 2, there are two competing grammars that are consistent with the English data available to the learner: an /ng/ → [ɥ] grammar and an /h/ → [ɥ] grammar. There are reasons to believe, however, that only the /ng/ → [ɥ] grammar is descriptively adequate. One example can be found in common slips of tongue, in which English speakers seem to pronounce the underlying /ng/: e.g., [swɪn] and [sweɪg] instead of [swɪŋ] and [sweɪ] ‘*swing and sway*’ (Kenstowicz and Kisseberth, 1977).

Another indirect evidence is mentioned by Chomsky and Halle (1968), who noticed that while there are words like ‘*line*’, ‘*lime*’, ‘*lane*’, and ‘*lame*’ in English, similar nonce words such as *[laɪŋ] and *[leɪŋ] are ungrammatical. According to Chomsky and Halle, this fact can be explained by the generalization that diphthongs in English cannot precede a sequence of consonants. This generalization can explain the ill-formedness of words such as [leɪŋ], in which [ɥ] follows a diphthong, as long as [ɥ] is derived from /ng/.

As will be shown imminently, under naive representational assumptions, SIMPLICITY favors the descriptively incorrect /h/ → [ɥ] grammar. This wrong prediction of SIMPLICITY constitutes a challenge from the perspective of explanatory adequacy that we are about to address.

The learning setting of generative linguistics was laid by Chomsky (1964, 1965) and is illustrated in (13). In this setting, the evaluation metric is a function that assigns a numerical score to each observationally adequate grammar in the child’s hypothesis space. The hypothesis with the best score is acquired.

(13) A schematic illustration of the learning setting (cf. McCarthy, 1981)



The original evaluation metric for phonological learning is the SPE metric (Chomsky and Halle, 1968). According to this metric, if G and G' can both generate the data D and G is simpler than G' , the learner should prefer G to G' . It has been claimed, however, that the SPE metric is not restrictive enough and therefore makes the wrong predictions when it comes to the acquisition of some phonological processes (more concretely, the acquisition of restricted optional processes, see Dell 1981).

MDL is an evaluation metric that follows the proposals of Solomonoff (1964), Kolmogorov (1965) and Chaitin (1966). MDL’s evaluation is determined by the economy of the grammar ($|G|$; cf. the SPE metric) but also by the economy of the description of the data given the grammar ($|D:G|$), which motivates restrictiveness. A formal definition of MDL is given in (14).

(14) MDL evaluation metric (Rasin et al., 2021):

If G and G' can both generate the data D , and if $|G| + |D:G| < |G'| + |D:G'|$, prefer G to G' .

MDL has shown success in supporting the acquisition of URs and morphological segmentation from distributional data, even in the face of notable learning challenges such as opacity and optionality. For a comparison of MDL with other phonological learners, see Rasin and Katzir (2016) and Rasin et al. (2021).

The $|D:G|$ part of MDL will not play a role in what follows because both competing grammars from section 2 are restrictive to the same extent as far as the calculation of $|D:G|$ is concerned.⁵ What matters to our discussion is only the cost of URs and phonological rules. Hence, we refer to both MDL and the SPE metric with the term SIMPLICITY that covers all simplicity-based evaluation metrics.

Turning to the challenge, as shown above, under naive representational assumptions, the wrong grammar has a cheaper lexicon (URs such as /sih/ which consist of strictly fewer segments than the equivalent /sing/) but more expensive rules (as it has an additional rule: $h \rightarrow \eta / _\sigma$). We will compare the grammars, translating their differences into a description length in bits, as schematically illustrated in (15). The grammar that can save more bits is favored by SIMPLICITY.

⁵Measuring $|D:G|$, however, is crucial for MDL’s success with respect to previous known challenges for the SPE metric, namely, the challenge noted by Dell (1981) from optional phonological processes (see Rasin and Katzir 2016; Rasin et al. 2021).

$$(15) \quad \underbrace{\underbrace{0101011010100\dots}_{\text{Lexicon}} \underbrace{10101010010\dots}_{\text{Rules}}}_{\text{G}}$$

In Rasin et al. (2021), a phonological rule costs a few dozen bits. Even if $h \rightarrow \eta$ requires substantially more bits than that, SIMPLICITY would favor the wrong grammar if its shorter URs make up a sufficiently cheaper lexicon. Therefore, in order to compare the grammars, we need to set the cost of an underlying segment. Per Rasin and Katzir (2020), the cost of a symbol in the lexicon is the amount of information required to identify one symbol from the set of possible symbols. The formula is given in (16) and the cost for English is calculated in (17).⁶

(16) The cost of a symbol in the lexicon
 $\log_2 x$, where x = number of symbols that can be used in the lexicon.

(17) The cost of an underlying segment in English⁷

- There are 51 underlying symbols in English. Therefore, x in English = 51.
- The cost of an underlying segment = $\log_2 x = \log_2 51 = 5.672$.

Note that the cost is relatively robust with respect to the exact number of underlying segments in English. Even if there are actually fewer underlying segments, the logarithmic function would result in a roughly similar cost for each segment (for example, $\log_2 40 = 5.322$). Such small differences are negligible in the calculation that we are about to see.

As mentioned before, the wrong grammar derives an $[\eta]$ in examples such as $/sih/ \rightarrow [si\eta]$ from a single segment, $/h/$. The correct grammar, on the other hand, derives such $[\eta]$'s from $/ng/$ sequences. Therefore, as determined by the calculation in (17), for every word in which an $[\eta]$ is derived from an underlying $/h/$ instead of $/ng/$, the wrong grammar requires 5.672 fewer bits (the cost of the additional segment). The difference in favor of the wrong grammar's lexicon is therefore $5.672 \cdot y$, where y = the number of words consisting of an $[\eta]$ derived from $/h/$ instead of $/ng/$. A summary of the relevant differences between the grammars is shown in (18).

⁶There are other conceivable ways to evaluate the cost of underlying representations, but we follow here Rasin and Katzir's (2020) naive segment-based choice. We will revise this choice in section 5, as part of our proposed solution.

⁷According to McMahon (2016), there are 50 underlying segments in English. However, McMahon includes an underlying $/\eta/$, which should not be considered as an underlying symbol in our study because we assume that $[\eta]$ is not part of the underlying inventory. We added morpheme and word boundaries to McMahon's count of segments because they also play a role in phonological processes and constitute symbols in the lexicon. Therefore, we set the number 51 for our calculation.

(18) The crucial differences between the grammars for SIMPLICITY

	/ng/ → [ŋ]	/h/ → [ŋ]
Difference in URs	/sing/, /sɒng/, etc.	/sih/, /sɒh/, etc.
Difference in rules		An additional rule: $h \rightarrow \eta / _\sigma$
Advantages in bits	A few dozens (estimated) for one fewer rule	$5.672 \cdot y$, y = the number of words consisting of an [ŋ] derived from /h/ instead of /ng/

To calculate the advantage of the /h/ → [ŋ] grammar, we have searched the Word List corpus of Python’s Natural Language Toolkit (NLTK; available online at www.nltk.org), which consists of 236,736 English words. We counted words that end with ‘ng’, excluding 5-letter or longer words that end with ‘ing’ (to exclude all verbal present progressive inflections). Our conservative search thus over-excluded relevant words like ‘bring’ and ‘spring’, and all mid-word coda [ŋ]’s (e.g., ‘England’). The search brought up 417 words, meaning that 2,365 bits can be saved by /h/ → [ŋ] and it is therefore favored over the descriptively correct grammar.

It should be noted that these 417 words include a few duplicated items (e.g., ‘king’ and ‘King’) and some words that are unlikely to be heard by an English learning child (e.g., ‘Palaung’ and ‘bettong’). However, even if a more reliable dataset would include just as little as 5% of these 417 words, SIMPLICITY would still favor the descriptively incorrect grammar, raising an explanatory adequacy challenge.

4 This is a general challenge

Before moving on to our proposed solution, it is important to briefly discuss some properties of the /h/ → [ŋ] challenge for phonological learning that make this challenge general in principle. This is because – in principle at least – there may be constraints that exclude the descriptively incorrect grammar from the child’s hypothesis space in the English case. For example, if one assumes a universal restriction that bans an /h/ → [ŋ] alternation, then the /h/ → [ŋ] grammar would be excluded from the child’s space. If this grammar is unavailable for the child, it would not be able to be selected by SIMPLICITY and therefore would not pose an explanatory adequacy challenge.⁸

However, we believe that this challenge requires a general solution, as a similar problem can arise in principle whenever the three conditions in (19) hold.

(19) General properties of the /h/ → [ŋ] challenge:

⁸As shown by Steriade (2023), a simple constraint that can ban an /h/ → [ŋ] alternation could be formalized using constraints along the lines of Zuraw (2007): *MAP(h,ŋ), which penalizes the mappings /h/ to [ŋ] and /ŋ/ to [h]. A universal hierarchy like the P-map (Steriade, 2001) may entail a domination of *MAP(h,ŋ) cross-linguistically and therefore rule out the /h/ → [ŋ] alternation. We will not develop such solutions in this paper for reasons explained shortly.

- a. A trigger of a phonological process is deleted (henceforth, *counterbleeding-by-deletion*, as in /ng/ → [ŋg] → [ŋ], where the trigger /g/ for the place assimilation of /n/ is deleted).
- b. The output segment of the counterbleeding-by-deletion interaction happens to be in complementary distribution with another segment in the language (like [ŋ] and [h]).
- c. There are enough instances of this interaction to outweigh an additional complexity of rules/constraints (similarly to our count of words with [ŋ] that can be derived from /h/).

The opaque interaction in (19a) is very common cross-linguistically. According to Kim’s (2024) survey of phonological opacity, counterbleeding-by-deletion constitutes almost half of all counterbleeding interactions (in 41 out of 87 instances of counterbleeding examined, a phonological process is counterbled by consonant deletion or vowel deletion). In principle, at least in some of them, (19b) and (19c) may hold as well.

One potential example of such a combination of properties can be found in Bulgarian. In Bulgarian, velar palatalization and yer-vowel deletion are involved in counterbleeding-by-deletion (Green 2004; Kim 2024). The first process turns /k/ into [tʃ] before front vowels and the second process deletes a yer vowel before a syllable containing a full vowel. The interaction of the processes is shown in the derivation of [martʃna] ‘dark.fem’ in (20).

(20) Counterbleeding-by-deletion in Bulgarian (cf. Green 2004)

	UR	/markǐna/
Velar palatalization		martʃǐna
Yer deletion		martʃna
	SR	[martʃna]

In [martʃna], the vowel that triggers palatalization, ǐ, is deleted and the palatalized velar tʃ surfaces before a sonorant consonant. Incidentally, according to Ignatova-Tzoneva (2019), d̂z never surfaces before sonorant consonants (l, r, m, n) in Bulgarian. Therefore, as shown in (21), in an alternative grammar of Bulgarian, [martʃna] can be derived from the UR /mard̂zna/, which consists of one fewer segment than the UR /markǐna/.

(21) A derivation of [martʃna] ‘dark.fem’ in an alternative grammar

	UR	/mard̂zna/
d̂z → tʃ / _C[+son]		martʃna
	SR	[martʃna]

Because velar palatalization and yer deletion are independently motivated, the d̂z → tʃ rule in (21) introduces an additional complexity into the alternative grammar. However, much like the /h/ → [ŋ] case, if there are roughly a dozen words in which [tʃ] can be derived from /d̂z/ instead of the longer /kǐ/, the addition of the extra rule would be justified from

the perspective of SIMPLICITY. It seems unlikely that the choice between the grammars in (20) and (21) is solely determined by such a small word count.

What makes the Bulgarian case study especially interesting is the proximity of $[\widehat{dz}] \sim [\widehat{tj}]$. When Twaddell (1935) claimed that no one could consent to regarding [h] and [ɣ] as variants of the same phoneme, he was presumably referring to the fundamental phonetic difference of the segments. As mentioned above, it is conceivable that some constraints on phonological mappings rule out the specific /h/ → [ɣ] alternation. However, even if some phonetically-based mechanism can solve the case for English, it is highly unlikely to do the same for Bulgarian.⁹

In addition to the Bulgarian example, it is possible to construct many other plausible languages which manifest the combination of properties in (19). Consider a hypothetical language in which underlying obstruents are only voiceless, as in Hawaiian (see Newbrand 1951; Lyovin et al. 1997/2017). This language has intervocalic voicing (e.g. /tap-a/ → [tabá]) and stressless high vowel deletion (e.g., /tani-la/ → [tanlá]) as systematic phonological processes (stress always falls on the final syllable). Finally, in this hypothetical language, as in English, [h] never occurs in coda position.

The generalizations about this language support an analysis according to which words like [tabmá] are derived from underlying /tapima/. However, under the naive segmental representations we have assumed so far, SIMPLICITY would prefer to derive [tabmá] from /tahma/ as long as there are enough words of this sort to support the addition of an /h/ → [b] rule to the grammar. Regardless of the exact number of words like [tabmá], this analysis does not seem plausible. Importantly, in this hypothetical example, no phonetic restriction is likely to rule out the incorrect grammar, because /h/ → [b] alternations are attested – for example, in Nganasan (see Vaysman 2009) – and therefore cannot be banned.

Since a phonetic solution seems unlikely, the solution we will pursue in section 5 is entirely formal, relying on information-theoretic considerations and having nothing to do with phonetic restrictions. As discussed in this section, we believe that the complementary-distribution challenge for phonological learning is general and can go beyond the particular case of [h] and [ɣ]. It is possible that a different mechanism can solve the problem for English, but any phonetic restriction will be an ad-hoc solution for a specific phonological pattern and not sufficient for the general case. In what follows, we will propose a solution that is not tailor made for any specific case study.

⁹One can argue against the possibility of the alternative Bulgarian grammar, on the grounds that no natural constraint can motivate pre-nasal devoicing. However, post-nasal devoicing and other phonological patterns are similarly unnatural and unmotivated, but nevertheless attested (see Beguš 2019). Such restrictions on phonological mappings may be too restrictive and rule out such attested patterns.

5 Approaching a solution: underspecification and lexical redundancy rules

As pointed out by Goodman (1943, 1949), simplicity is defined relative to a representational framework. Consider, for example, two alternative representations of numbers: the Arabic numeral system (0, 1, 2, 3, 4, 5, 6, 7, 8, 9) and the Roman numeral system (I, V, X, L, C, D, M). As shown in (22), if numbers are represented using Arabic numerals, according to a metric that counts letters, thirty-eight (38) is simpler than a thousand (1000). However, if numbers are represented using Roman numerals, the picture changes and a thousand (M) is simpler than thirty-eight (XXXVIII).

(22) Simplicity of numbers is defined relative to a numeric system

Num. system	One thousand	Thirty-eight
Arabic	1000	38
Roman	M	XXXVIII

In linguistics as well, different theories of representation will make different predictions about acquisition when combined with a simplicity-based learning criterion. This makes it possible to reason about the correct representational system based on what is acquired using SIMPLICITY. Early work in generative linguistics used this kind of argumentation (Halle, 1962; Chomsky, 1965). Recently, this methodology has been revived using MDL (Katzir, 2014; Katzir et al., 2020; Rasin and Katzir, 2020). In line with this work, the present paper proposes to rethink the /h/ → [ŋ] puzzle as an argument for specific representations rather than an argument against SIMPLICITY.

Halle (1959, 1962) and Chomsky and Halle (1968) proposed that the lexicon consists of an underspecified level of representation that is filled by lexical redundancy rules. In what follows we will show that these representational assumptions, which are given in (5) and repeated in (23), can solve the explanatory adequacy challenge discussed in this paper.

(23) Representational assumptions that can solve the challenge

1. Underlying segments can be underspecified for phonological features as long as they are pairwise distinct in every environment.
2. Unspecified features are filled by lexical redundancy rules.

Combined with the assumptions in (23), which will be explained immediately, SIMPLICITY favors grammars in which segments are *minimally specified*, given the *minimal set of features* that can generate the data.

The assumptions in (23) follow from Halle (1959, 1962) and Chomsky and Halle (1968), according to which children try to minimize the number of features they store in their lexicon and can add a lexical redundancy rule to their grammar that fills in the missing features. However, they only do so when adding such a rule to the grammar leads to savings in the lexicon that outweigh the cost of the rule.

Halle (1962, p. 60) and Chomsky and Halle (1968, p. 381) made their argument using the following example. English speakers judge the nonce word *[vnig] as unacceptable but the nonce word [bɪk] as acceptable. For the first word, there is a generalization over the English lexicon that if the second consonant in a word-initial cluster is a nasal or an obstruent, the first consonant must be /s/. The predictability of /s/ in words such as ‘snake’ and ‘skin’ allows storing the first consonant of those words simply as [+consonantal] and filling in the remaining features of /s/ by a rule like (24), thus saving multiple features in the lexical representation of many words in the lexicon.

(24) [+consonantal] → [+strident, -voice, etc.] / # __ [+consonantal, etc.]

The rule in (24) bans *[vnig] (and many other ungrammatical nonce words like *[tsajm], *[kbait], etc.) because it requires the consonant preceding /n/ to be /s/.

Unlike *[vnig], [bɪk] differs minimally from English words like ‘pick’, ‘buck’, and ‘big’. Therefore, a rule that bans [bɪk] but not those words is limited in the number of features it can reduce from the lexicon. Consider, for example, a rule stating that a dorsal stop (/k/ or /g/) in the context [#bɪ_#] must be voiced (/g/). This rule could only save one feature, [+voice], from the lexical entry of ‘big’, and therefore, according to Chomsky and Halle, will not be acquired.

Since Halle (1959, 1962) and Chomsky and Halle (1968), a lot of ink had been spilled on underspecification (see, for example, Kiparsky 1982, 1993; Archangeli 1988; Steriade 1995) and, along the paradigm shift to OT, theoretical phonology had shifted away from Chomsky and Halle’s original proposal. In what follows, however, we show that under the original theory of underspecification, the correct /ng/ → [ŋ] grammar is actually the simpler option, and thus preferred by SIMPLICITY.¹⁰

The preference of SIMPLICITY for the correct /ng/ → [ŋ] grammar is motivated by a generalization that can only be stated in that grammar and can be used to simplify it:

(25) There are no /h/’s in coda position across the English lexicon.

While the correct /ng/ → [ŋ] grammar is consistent with this generalization, in the incorrect /h/ → [ŋ] grammar, there must be underlying /h/’s in codas to derive words like ‘sing’ and ‘song’ (e.g., /sɔh/ → [sɔŋ]).

To understand how the assumptions in (23) allow incorporating (25) into the grammar, consider first the toy feature tables in (26) and (27). In order to distinguish between three segments, /n/, /g/, and /k/, as shown in (26), it is possible to use just two features (here chosen to be [±sonorant] and [±voice] for demonstration). However, once we also add /h/ to the inventory, as shown in (27), /k/ is no longer distinguishable by those two features alone. To distinguish between /k/ and /h/, /k/ must be specified for an additional feature (here chosen to be [±continuant]).

¹⁰It is reasonable that different theories of underspecification can also solve this challenge, as it is possible that others do not. The comparison between theories of underspecification goes beyond the scope of this paper.

(26) Minimally-specified /n/, /g/,
and /k/ (without /h/)

	n	g	k
±sonorant	+	-	-
±voice	+	+	-
±continuant			

(27) Minimally-specified /n/, /g/,
/k/, and /h/

	n	g	k	h
±sonorant	+	-	-	-
±voice	+	+	-	-
±continuant			-	+

According to the first assumption in (23), segments must be pairwise distinct. i.e., every segment must have a unique specification that distinguishes it from every other segment (a subset of another specification does not constitute a unique specification). This is required for the application of lexical redundancy rules (see Chomsky and Halle 1968, pp. 382-389 for further discussion). Consequently, if /h/, which is [-sonorant, -voice, +continuant], is added to the underlying inventory, /k/, which is also [-sonorant, -voice] must be specified as [-continuant] to be distinguishable from /h/.

Therefore, if underspecification is allowed and there are no /h/'s in coda position, as in the correct grammar, the exclusion of underlying /h/'s from coda position may lead to a mass reduction of features from the lexicon. Because of the exclusion of coda /h/'s, the features that distinguish between /k/ and /h/ can be removed from coda /k/'s in the lexicon and filled in by a lexical redundancy rule.

Under this architecture, we can now revise the calculation in (18) and count features in the lexicon instead of segments. For a concrete evaluation, consider now the exhaustive feature table in (28). The table shows all fully-specified underlying English consonants, assuming the SPE theory of features.^{11,12}

(28) Fully-specified English consonants

	p	b	f	v	m	t	d	θ	ð	s	z	ʃ	ʒ	tʃ	dʒ	n	r	l	k	g	j	w	h
±son	-	-	-	-	+	-	-	-	-	-	-	-	-	-	-	+	+	+	-	-	+	+	-
±voc	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	+	+	-	-	-	-	-
±cons	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	-	-	-
±cor	-	-	-	-	-	+	+	+	+	+	+	+	+	+	+	+	+	+	-	-	-	-	-
±ant	+	+	+	+	+	+	+	+	+	+	+	-	-	-	-	+	-	+	-	-	-	-	-
±voice	-	+	-	+	+	-	+	-	+	-	+	-	+	-	+	+	+	+	-	+	+	+	-
±cont	-	-	+	+	-	-	-	+	+	+	+	+	+	-	-	-	+	+	-	-	+	+	+
±nasal	-	-	-	-	+	-	-	-	-	-	-	-	-	-	-	+	-	-	-	-	-	-	-
±str	-	-	+	+	-	-	-	-	-	+	+	+	+	+	+	-	-	-	-	-	-	-	-
±back	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	+	+	-	+	-
±high	-	-	-	-	-	-	-	-	-	-	-	+	+	+	+	-	-	-	+	+	+	+	-
±low	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	+

¹¹We use some common abbreviations here and below: son = sonorant, voc = vocalic, cons = consonantal, cor = coronal, ant = anterior, cont = continuant, and str = strident.

¹²Chomsky and Halle (1968) also use the features [±tense] which is unspecified for all consonants and [±round] which is unspecified for all consonants in (28) except for /k/ and /g/ (both are [-round]).

The minimally-specified version of the table in (28) is shown in (30). The table was constructed while preserving Rasin and Katzir’s (2020) assumption about cost, given in (16) and repeated in (29).

(29) The cost of a symbol in the lexicon

$\log_2 x$, where x = number of symbols that can be used in the lexicon.

Henceforth, $\log_2 x = c$

Just like the cost formula in (29) created pressure for reducing the number of underlying segments in our previous segment-based calculation, it motivates the minimization of the number of features used in the lexicon under a feature-based calculation. The table in (30) therefore shows the minimal specification of English consonants given the minimal set of distinctive features.

(30) Minimally-specified English consonants

	p	b	f	v	m	t	d	θ	ð	s	z	ʃ	ʒ	tʃ	dʒ	n	r	l	k	g	j	w	h
±son	-	-			+			-	-	-	-	-	-			+	+	+	-	-	+	+	-
±cor	-	-	-	-	-	+	+	+	+	+	+	+	+	+	+	+	+	+	-	-	-	-	-
±ant	+	+	+	+		+	+	+	+	+	+	-	-	-	-	+	-	+	-	-	-	-	-
±voice	-	+	-	+		-	+	-	+	-	+	-	+	-	+				-	+			-
±cont	-	-	+	+	-	-	-	+	+	+	+	+	+	-	-	-	+	+	-				+
±str								-	-	+	+												
±back																					-	+	

The table in (30) is an approximation of the simplest way to specify the consonants of English under the pressure to minimize the general set of features.¹³ For example, /m/ and /n/ could have been defined using [+nasal], but they are specified using [+son, -cont] because [±son] and [±cont] are independently required. A feature is only used if it is necessary. An example is the feature [±str], which is needed to distinguish between the coronal fricatives /θ/, /ð/ and /s/, /z/. After constructing the table using the minimal set of distinctive features, a secondary pressure is to reduce every redundant feature from the specifications of every segment in the lexicon. Consider /k/ for example: [-son] is required to distinguish it from /j/ and /w/, [-cor] from /tʃ/, [-ant] from /p/, [-voice] from /g/, and [-cont] from /h/. All other features of /k/ are redundant and can be filled in by lexical redundancy rules.

However, in the correct /ng/ → [ŋ] grammar, there are no /h/’s in codas. The feature [-cont] can be removed from the specification of /k/ in coda position, because there is no other [-son, -cor, -ant, -voice] consonant in the same position in that grammar. This is visualized in (31).

¹³The table makes certain arbitrary choices for concreteness. For example, /j/ and /w/ are specified as [-ant] in order for them to be distinguished from /f/ and /v/. Alternatively, we could have added the specification of [-son] to /f/ and /v/. Such a decision can affect the cost of the entire lexicon, given the frequency of the segments. SIMPLICITY could thus favor one option over the other. In this study, however, we focus on the feature count for the segments /n/, /g/, /h/, and /k/, which are the segments crucial to us.

(31) Minimally-specified English consonants without /h/	p	b	f	v	m	t	d	θ	ð	s	z	ʃ	ʒ	tʃ	dʒ	n	r	l	k	g	j	w
±son	-	-			+			-	-	-	-	-	-			+	+	+	-	-	+	+
±cor	-	-	-	-	-	+	+	+	+	+	+	+	+	+	+	+	+	+	-	-	-	-
±ant	+	+	+	+		+	+	+	+	+	+	-	-	-	-	+	-	+	-	-	-	-
±voice	-	+	-	+		-	+	-	+	-	+	-	+	-	+				-	+		
±cont	-	-	+	+	-	-	-	+	+	+	+	+	+	-	-	-	+	+				
±str								-	-	+	+											
±back																					-	+

The crucial difference between (30) and (31) was highlighted in (3) and (4) and is repeated here in (32) and (33). In the incorrect /h/ to [ŋ] grammar, the lexicon must include enough features to distinguish between underlying /h/ and /k/ in coda position. As shown in (32), the feature [±continuant] is required to make /k/ distinguishable from /h/. In the correct grammar, however, /h/ never appears in coda position, so it is redundant to specify the feature [-continuant] for coda /k/, as illustrated in (33).

(32) Underspecified segments in coda position, **incorrect grammar**

	p	b	...	k	h
±coronal	-	-	...	-	-
±anterior	+	+	...	-	-
...			...		
±voice	-	+	...	-	-
±continuant	-	-	...	-	+

(33) Underspecified segments in coda position, **correct grammar**

	p	b	...	k
±coronal	-	-	...	-
±anterior	+	+	...	-
...			...	
±voice	-	+	...	-
±continuant	-	-	...	

We now proceed to the actual evaluation of the competing grammars. Unlike the calculation in (18), under the current feature-based assumptions, the cost of an underlying segment is determined by the number of features required to specify it in (30) or (31). The crucial differences in the representation of the grammars are shown in (34), before we proceed to calculate the cost of features using bits.

(34) A revision of (18), assuming underspecification.

	/ng/ → [ŋ]	/h/ → [ŋ]
Difference in URs	/sing/, /sɔŋ/, etc.	/sih/, /sɔh/, etc.
Difference in rules		/h/ → ŋ / __]σ
Advantages	The additional feature required to distinguish between /k/ and /h/, for every coda /k/.	The additional features required to state /ng/ instead of /h/, for every [ŋ].

Note that the lexical redundancy rules themselves do not play a role in this comparison, because the only difference between these grammars in this respect is in the statement of

the feature-filling rule for /k/ and, as shown in (35), it has the same complexity in both grammars. The only difference between the two versions of the rule is that the feature [-cont] is stated in the output of the rule in the /ng/ to [ŋ] grammar, but in the input of the rule in the /h/ to [ɥ] grammar.

(35) Feature-filling rule for /k/ in coda position:

- /ng/ → [ŋ] grammar:
[-son, -cor, -ant, -voice] → [-**cont**, -str, etc.] / $__\sigma$
- /h/ → [ɥ] grammar:
[-son, -cor, -ant, -voice, **-cont**] → [-str, etc.] / $__\sigma$

Given the featural representations in (30) and (31), storing /sih/ instead of /sing/ in the lexicon of the /h/ → [ɥ] grammar saves the cost of 7 symbols (3 additional features, their binary value, and an additional separator of segments, which is required to divide the different bundles of features that specify /n/ and /g/). On the other hand, excluding /h/ from codas in the correct grammar saves 2 symbols (the feature [$\pm$ continuant] and its value) for every /k/ in coda position. Because we only counted word-final [ŋ]’s in section 3, we also count here only words that end with ‘k’, excluding many medial-word coda /k/’s (e.g., [síkɾit] ‘*secret*'). We searched the same corpus of Python’s NLTK package and found 2,674 such words. The new evaluation is shown in (36).

(36) A revision of (18), assuming underspecification - **bit calculation**

	/ng/ → [ŋ]	/h/ → [ɥ]
Difference in URs	/sing/, /sɔng/, etc.	/sih/, /sɔh/, etc.
Difference in rules		/h/ → ɥ / $__\sigma$
Advantages in bits	A few dozens (estimated) for one fewer rule. 2 symbols for every /k/ in coda $= 2 \cdot 2,674 \cdot c = 5,348 \cdot c$	7 symbols for each word with [ɥ] derived from /h/ instead of /ng/ $= 7 \cdot 417 \cdot c = 2,919 \cdot c$

As can be seen in (36), the /ng/ → [ŋ] grammar now wins. However, what may still be worrying in the picture portrayed by (36) is the fact that if there were sufficiently more than 417 relevant words in the /h/ → [ɥ] grammar, the result would have been reversed. In section 4, we argued that the logical analysis should not be determined by a small word count. A closer examination of the current situation reassures that this is indeed not the case here.

Consider the conditions in which, assuming underspecification, the /ng/ → [ŋ] grammar would be disfavored. In order for the advantages of the /h/ → [ɥ] to outweigh the correct grammar, the number of words with [ɥ] derived from /h/ must be almost double than the current 417. Roughly 775 words would have been enough to justify the addition of a feature to coda /k/’s and the /h/ → [ɥ] rule. First of all, this is not a small modification, but a substantial change to the lexicon. However, more importantly, given the representational

assumptions in (36), there is no reason to assume that the change that the addition of hundreds of words will bring to English will be the derivation of [ŋ] from /h/. Currently, there is no clear reason to add /h/’s to coda position instead of underlying /ŋ/’s. In the previous story, both the /ng/ → [ŋ] and the /h/ → [ŋ] were simpler than the /ŋ/ → [ŋ] grammar, but the current situation is different.

Under the naive assumptions that we have started our paper with, the main simplicity-based pressure was to reduce the number of underlying segments and therefore both /h/ → [ŋ] and /ng/ → [ŋ] were favored over /ŋ/ → [ŋ]. This was because an addition of a segment to the underlying inventory was translated to an additional cost for every segment in every lexical item (because the cost of a segment was determined by the number of underlying segments). Given the new properties, however, adding /ŋ/ to coda positions is much less expensive. In the current state of affairs, adding distinctive features to the underlying inventory inflates the entire lexicon. However, as long as adding a segment to a specific environment does not require the expansion of the general set of features, such an addition only bears on the specification of one other segment in one specific position. Much like adding /h/ to coda positions above led to the addition of a feature for coda /k/’s.

Indeed, adding /ŋ/ to the lexicon does not require more features than those used in (30) and (31). Adding /ŋ/ to coda positions would potentially bear on the specification of coda /m/’s (the only other [+son, -cor, -cont]) but not on other segments.¹⁴ Moreover, deriving [ŋ] from /ŋ/ also does not involve the cost of the additional phonological rule in the /h/ → [ŋ] grammar. A small addition of words will therefore not lead to a simplicity-based learner’s preference for the /h/ → [ŋ] grammar. A massive addition of words, however, can result in a switch to a /ŋ/ → [ŋ] grammar. The prediction, which seems like a desirable prediction, is therefore that the frequency of words with [ŋ] could influence the choice between the /ng/ → [ŋ] and the /ŋ/ → [ŋ] grammars.

To summarize this section, under the representational assumptions of Halle (1959, 1962) and Chomsky and Halle (1968), SIMPLICITY favors the descriptively correct grammar of English over the /h/ → [ŋ] grammar. In the correct grammar, as outlined in section 2, [ŋ] is derived from the sequence /ng/ in coda position. /n/ assimilates to /g/, which is deleted by process that targets non-coronal voiced stops after nasals (e.g., /sing/ → [sɪŋɡ] → [sɪŋ]). SIMPLICITY chooses this grammar even though it has longer URs, because this grammar captures the generalization in (25), which can simplify the lexicon elsewhere. The exclusion of /h/ from coda position, assuming underspecification, licenses the removal of the feature [-cont] from all coda /k/’s. This feature is filled in by the lexical redundancy rule given in (35) and repeated here: [-son, -cor, -ant, -voice] → [-cont, -str, etc.] / __]σ.

¹⁴Given the table in (31), in order to be distinguishable from /ŋ/, /m/ needs the additional specification of [+ant], as /ŋ/ can be specified as [+son, -cor, -ant, -cont].

6 Conclusion

The complementary distribution of /h/ and [ŋ] was a challenge for structuralism. As we saw, this distribution also poses a challenge for simplicity-based learning in generative linguistics: under naive representational assumptions, the descriptively incorrect /h/ → [ŋ] grammar constitutes a simpler option than the descriptively correct /ng/ → [ŋ] grammar. A simplicity-based learner therefore favors the incorrect grammar, raising an explanatory adequacy challenge.

Chomsky and Halle (1968) did not seem to worry about this challenge, even though it dates back to the SPE metric. It is conceivable that they were not aware of this challenge, but it is also possible that they did not consider it to be a challenge because they had already posited the representational assumptions that eliminate it. Either way, this problem could have been set aside for a while because simplicity-based learning was abandoned. Simplicity-based learning, however, has recently returned in the form of MDL, which can support the acquisition of phonological grammars, as discussed in section 3. Given that MDL makes SIMPLICITY central again as a principle for phonological learning, the challenge of the complementary distribution of /h/ and [ŋ] can no longer be ignored.

The solution we proposed here was to rethink the /h/ → [ŋ] challenge, not as a challenge for SIMPLICITY per se, but as a challenge for a combination of SIMPLICITY and a particular representational assumption – the assumption that underlying segments are fully specified. If we adopt different representational assumptions, the problem disappears. Whether they were aware of it or not, Chomsky and Halle did not have to worry about /h/ → [ŋ] because they already had all of the ingredients that can solve the challenge: underspecification and lexical redundancy rules.

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